MATE664 L20

Continuum modelling & phase-field

Generalized Fick's equation

$$\boxed{\frac{\partial \xi}{\partial t} = -\nabla \cdot \vec{J}(f, \xi)}$$

ξ: extensive quantity (conserved)

f: free energy functional

- When dealing with diluted solution or mixture system, we can use f of macroscopic free energy term
- If phase boundaries exist, must account for interfacial terms
- Usually solved on discretized grids (FD or FEM FVM)
  1D, 2D & 3D

A general format is $J = -M \cdot \nabla F$

F: driving force as chemical potential

$$F = \left[\frac{\partial f}{\partial c} - \kappa \nabla^2 c \right]$$

$\frac{\partial f}{\partial c}$: homogeneous

$\kappa \nabla^2 c$: gradient term in Cahn-Hilliard

κ: gradient "penalty", not curvature

See example in lecture notes & assignment 4 for simulation details.

Goal $\Rightarrow$
- $\frac{\partial^2 f}{\partial c^2} < 0 \Rightarrow$ instability with length scale
- $\frac{\partial^2 f}{\partial c^2} > 0 \Rightarrow$ normal diffusion, flattens c-profile

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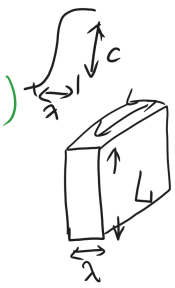
How to correlate parameter κ with γ ?

(Ref. MIT course 10.626 Electrochemical Energy Systems)

Use scaling analysis assume interface thickness λ

Energy density $\sim \kappa (\nabla c)^2 \sim \kappa \left(\frac{c}{\lambda}\right)^2 \sim w$

(energy scale for miscibility gap)



Also assume a "wall" of length L

Interfacial energy $\sim w \cdot \lambda \cdot L^2$
$\sim \gamma \cdot L^2$

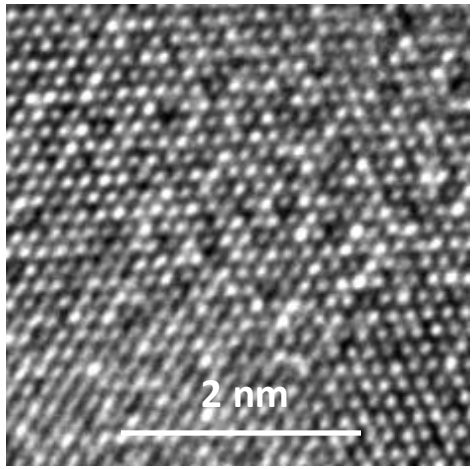
$\Rightarrow \gamma \sim w \cdot \lambda \sim \kappa \frac{c^2}{\lambda}$

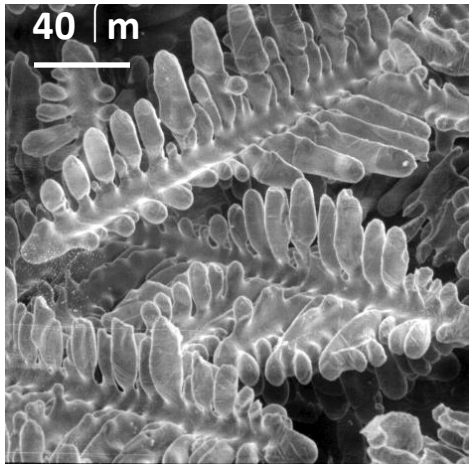
This is a powerful scaling relation, meaning any interface problem can be either modelled by either
- a sharp interface + γ
- diffuse interface + (λ, κ)

↳ Basics of phase-field simulation

The following handouts are adapted from MATE374 by Prof. Hao Zhang

Modeling at Various Length Scales





Atomistic scale Å

Dendrite scale μm

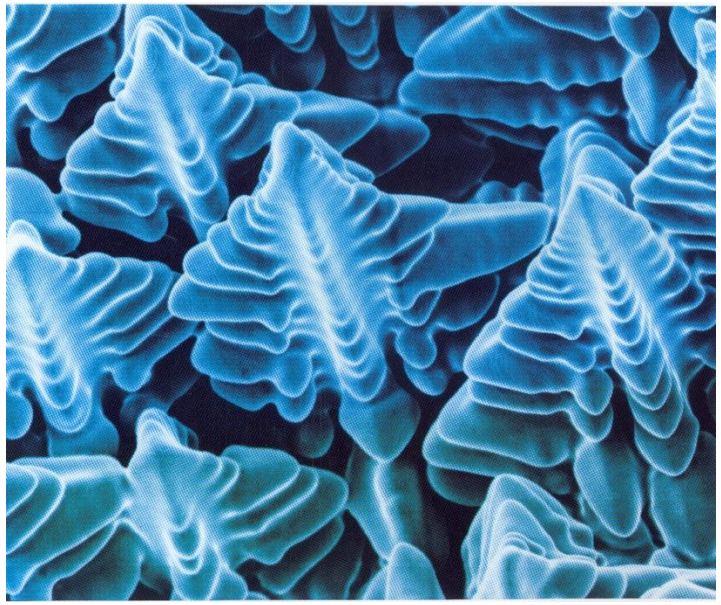
Grain scale mm

How to connect these various scales ?

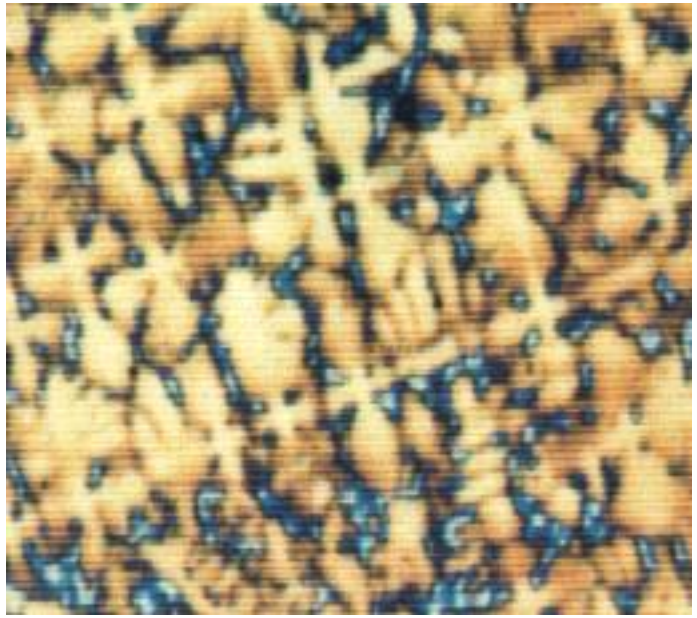
Component scale
cm - m

M. Rappaz, EPFL

Dendritic Microstructure

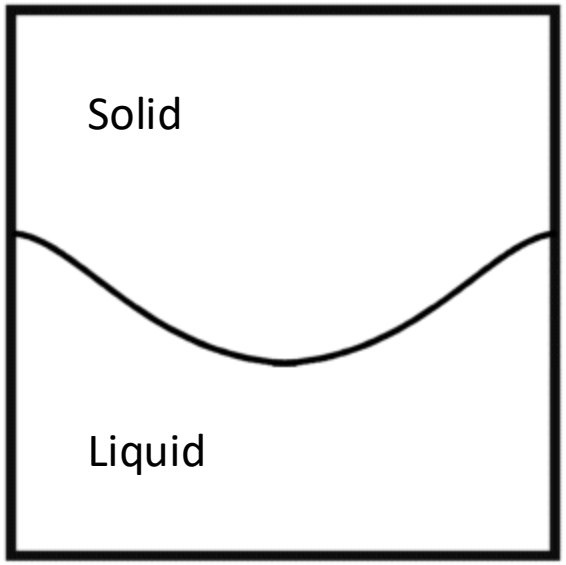


Liquid decanted during freezing



Polished and etched microstructure after freezing

Stefan Problem



- Interface is a surface;
- No thickness;
- Physical properties:
  - Surface energy, kinetics

$$T_S = T_L = T_M - T_M \frac{\gamma}{L_V} \mathcal{K} - v_n/\mu$$

  - Conservation of energy

$$k \frac{\partial T}{\partial n}\bigg|_L^S = L_V v_n$$

Surface Energy

- Critical Nucleus and Coarsening

  Critical Nucleus: $R_c = \dfrac{2T_M\gamma}{L_V(T_M - T)}$

  Coarsening: Minimize the total surface energy for a given volume of inclusions

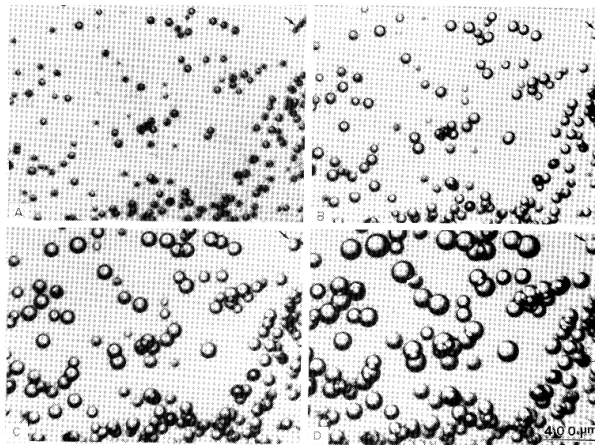
- Grain Boundary Grooves

$$\left(\frac{T_M\gamma}{L_V}\right)\frac{y_{xx}}{[1+y_x^2]^{3/2}} = -Gy$$

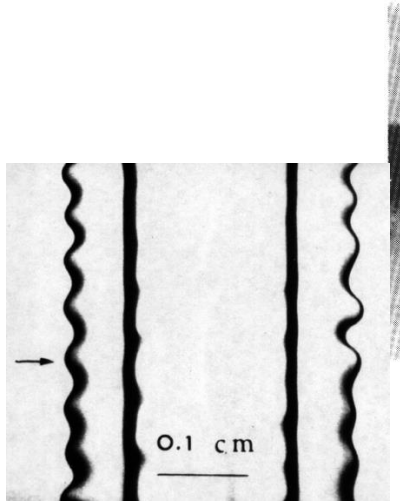
- Wavelength of instabilities

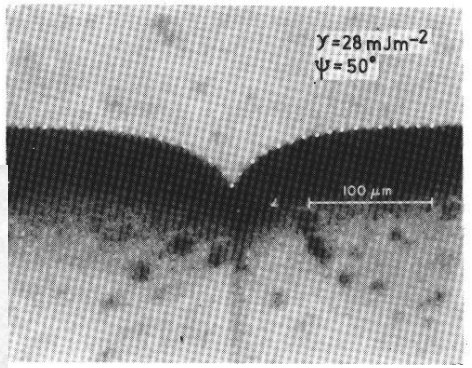
Ice cylinder growing into supercooled water, $T < T_M$

Instability wavelength depends on surface energy: $\lambda \sim \gamma^{1/2}$



P. Voorhees & R. Schaefer (1987)





S. Hardy and S. Coriell (1968)

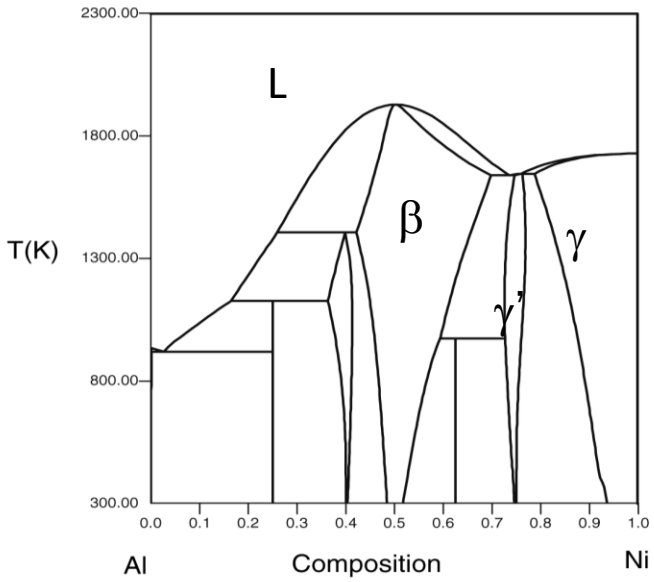
Phase Field Models

The phase-field model was developed around 1978 by J. Langer at CMU as a computational technique to solve Stefan problems for a pure material.

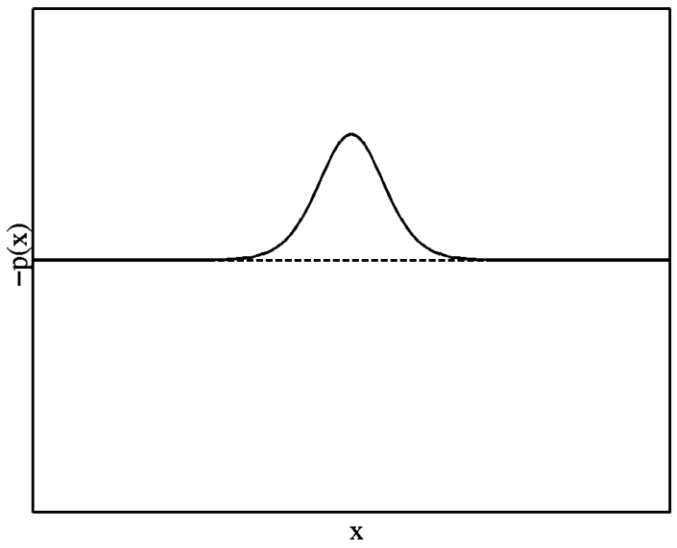
Main idea: Solve a single set of PDEs over the entire domain

Two main issues for a phase-field model:

Bulk Thermodynamics

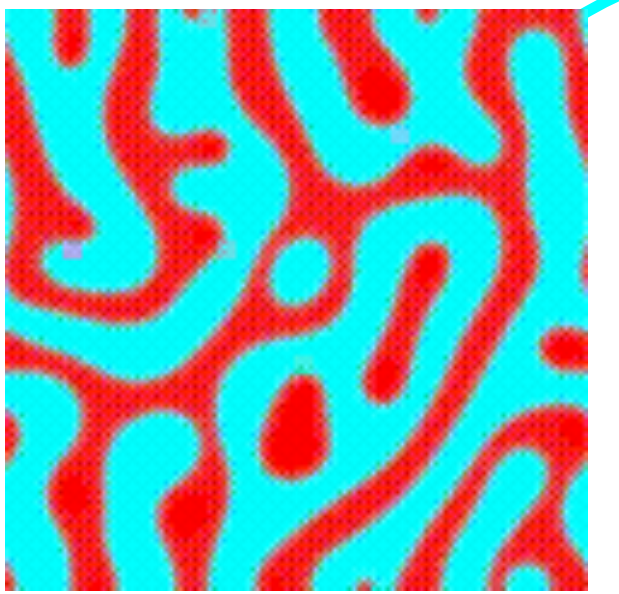


Surface Thermodynamics

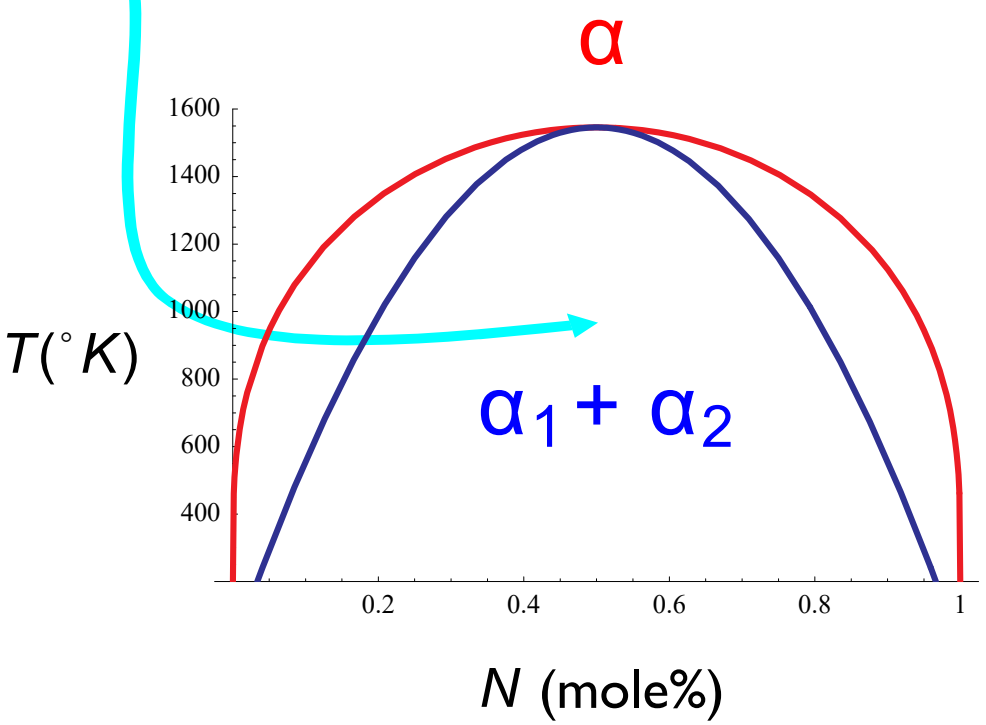


Phase-field model incorporates both bulk thermodynamics of multiphase systems and surface thermodynamics (e.g., Gibbs surface excess quantities).

Spinodal Decomposition



T_c = 1520K



Ω = 25.7kJ/mole

R. Balluffi, C. Carter, S. Allen "Kinetics of Materials" Wiley 2006

The Cahn-Hilliard Equation

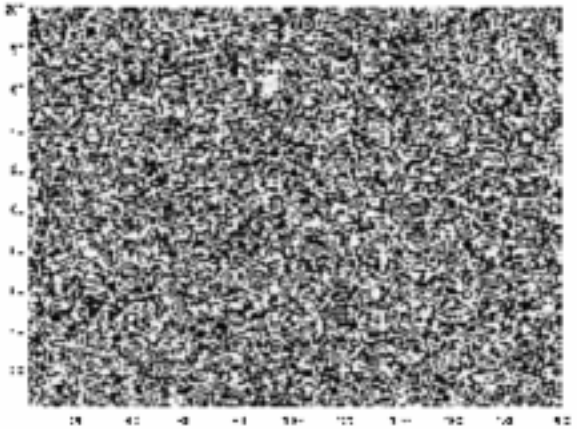
(for conserved quantities)

chemical

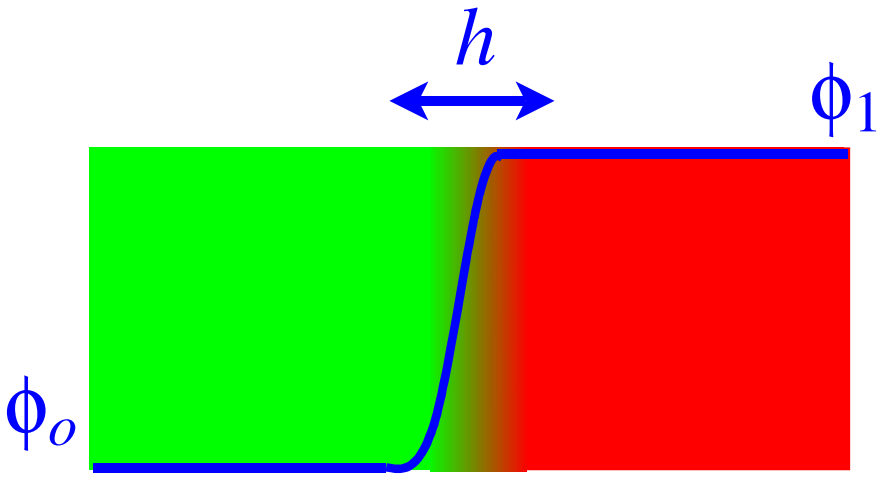
$$F[\phi, T, \ldots] = \int_\Omega f[\phi, T, \ldots] + \frac{k}{2}(\nabla\phi)^2 d\Omega$$

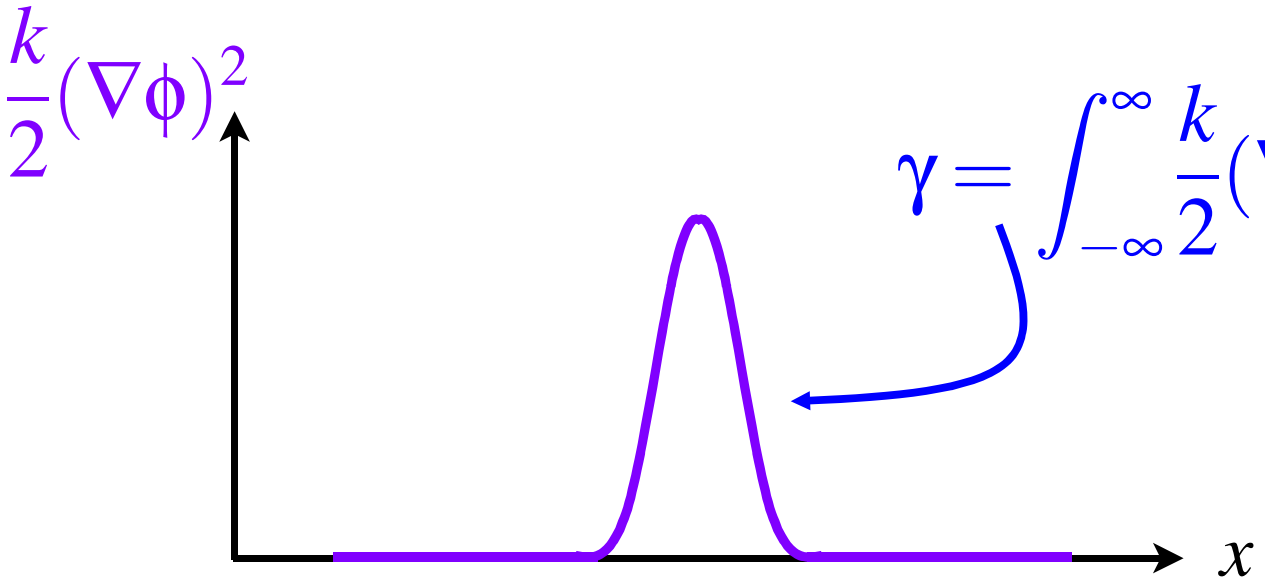
$$\frac{\partial\phi}{\partial t} = -\nabla \cdot M\nabla\left(\frac{\delta F}{\delta\phi}\right)$$

$$\boxed{\frac{\partial\phi}{\partial t} = M\left(\nabla^2\frac{\partial f}{\partial\phi} - k\nabla^4\phi\right)}$$



Diffusive Interface Model





$$\gamma = \int_{-\infty}^{\infty} \frac{k}{2}(\nabla\phi)^2 dx$$

Conceptual Implications

- Equilibrium is reached when every volume elements of materials stops evolving in time
- Phase separation can be represented by a PDE that includes the (measurable) thermodynamic properties of the system
- Interfacial effects are includes
- The question remains: Can this be applied to the generality of phase transformations?

Phase Field Model

- Introduce the phase-field variable:

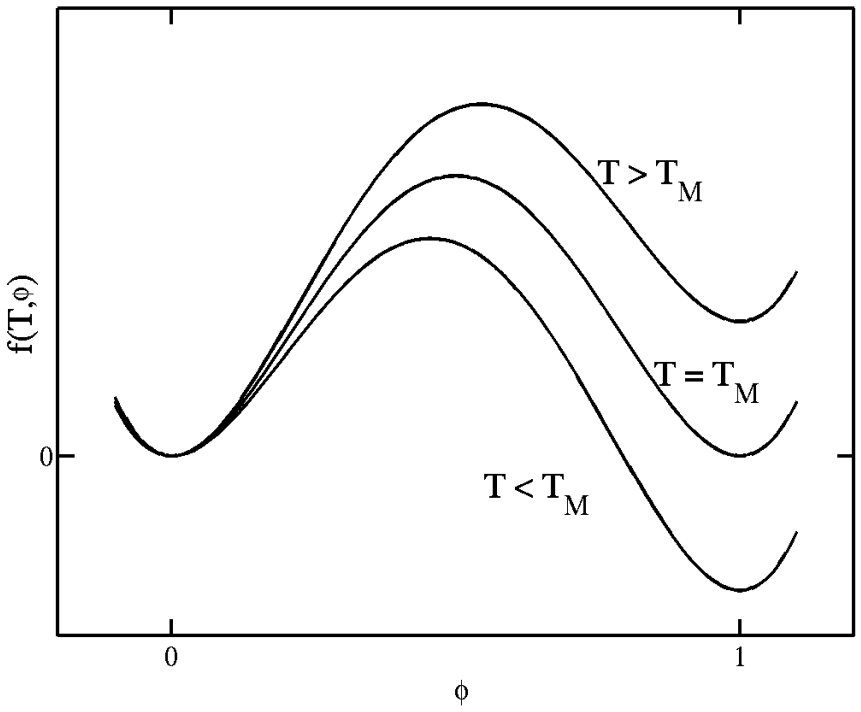
$$\text{Solid: } \phi(x,t) = 1$$

$$\text{Liquid: } \phi(x,t) = 0$$

- Introduce free-energy functional:

$$F = \int_V \left[\frac{1}{2}\varepsilon^2 |\nabla\phi|^2 + f(\phi,T) \right]$$

J.S. Langer (1978)



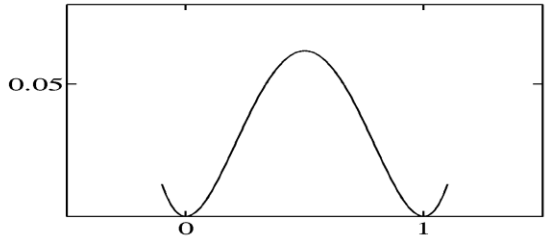
- Dynamics

$$\frac{\partial\phi}{\partial t} = -M\frac{\delta F}{\delta\phi}$$

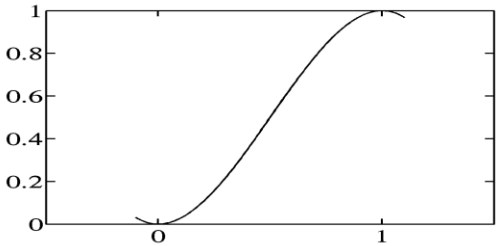
Free Energy Function

$$f(\phi,T)=\frac{W}{2}g(\phi)+\frac{L_V(T-T_M)}{T_M}p(\phi)$$

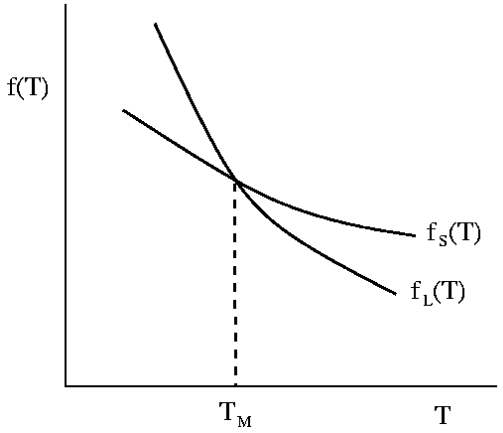
$$g(\phi)=\phi^2(1-\phi^2)$$



$$p(\phi)=\phi^2(3-2\phi)$$



$$f_S(T)-f_L(T)=f(1,T)-f(0,T)=\frac{L_V(T-T_M)}{T_M}$$



- Phase-field models provide a regularized version of Stefan problems for computational purposes
- Phase-field models are able to incorporate both bulk and surface thermodynamics
- Can be generalised to:
  - include material deformation (fluid flow & elasticity)
  - models of complex alloys
- Computations:
  - provides a vehicle for computing complex realistic microstructure

- Advantages - no need to track interface
  - can compute complex interface shapes

- Disadvantage - have to resolve thin interfacial layers

Phase Field Software

- Parallel Algorithms for Crystal Evolution in 3D (PACE3D) is a parallelized phase-field simulation package including multi-phase multi-component transformations, large scale grain structures and coupling with fluid flow, elastic, plastic and magnetic interactions. It is developed at the Karlsruhe University of Applied Sciences and Karlsruhe Institute of Technology.

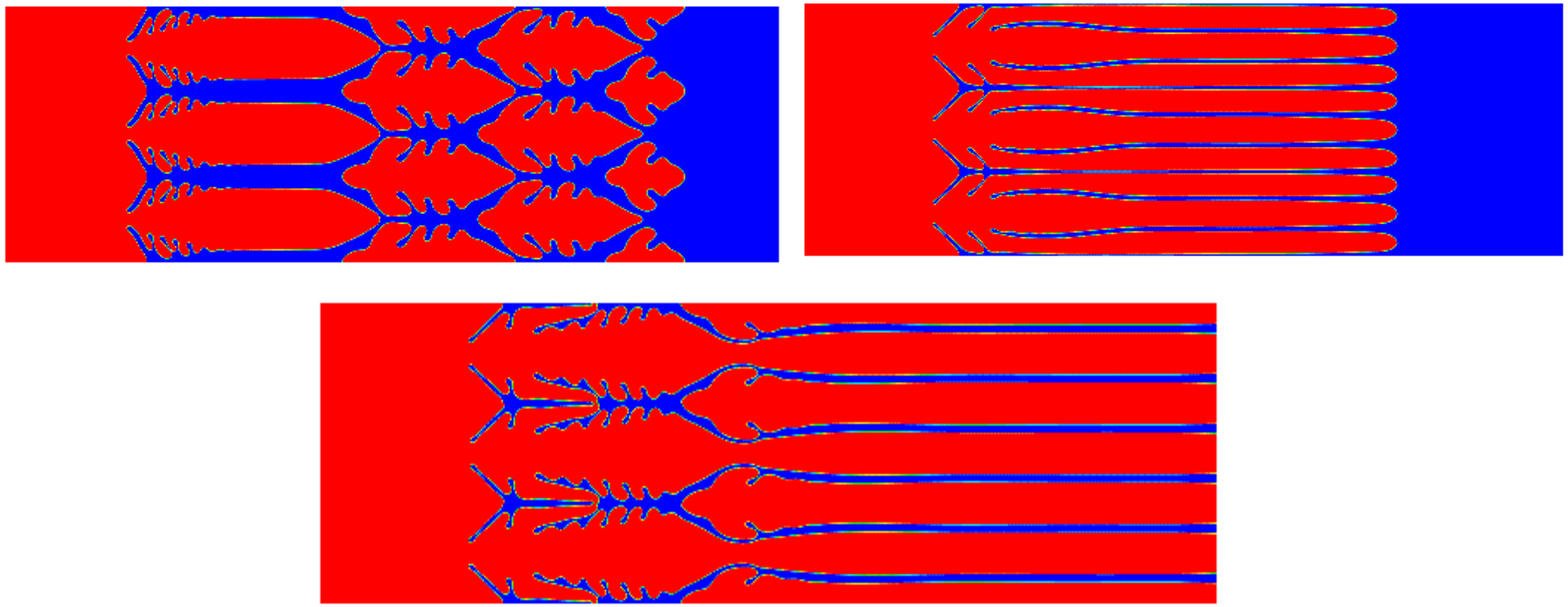
- The Microstructure Evolution Simulation Software (MICRESS) is a multi-phase field simulation package developed at RWTH-Aachen

- MOOSE massively parallel open source C++ multiphysics finite element framework with support for phase field simulations developed at Idoho National Laboratory.

- PhasePot is a Windows-based microstructure simulation tool, using a combination of phase field and Monte Carlo Potts models

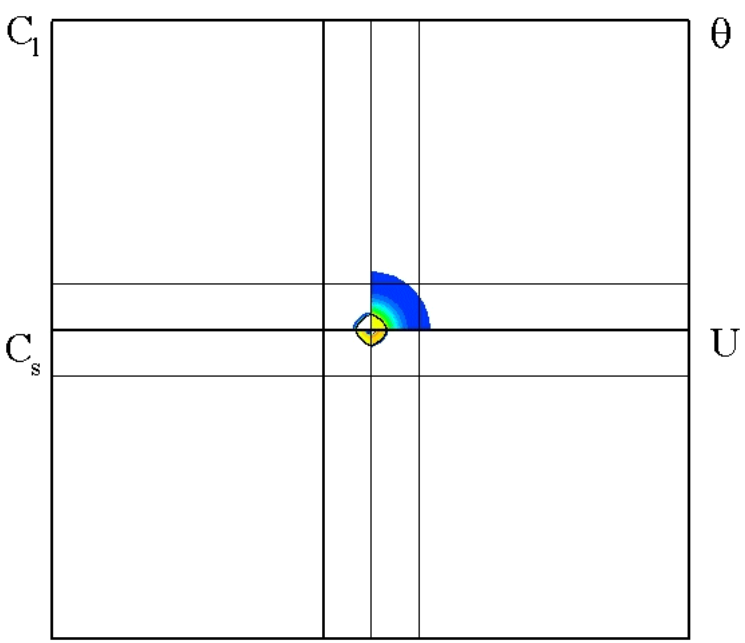
http://en.wikipedia.org/wiki/Phase_field_models

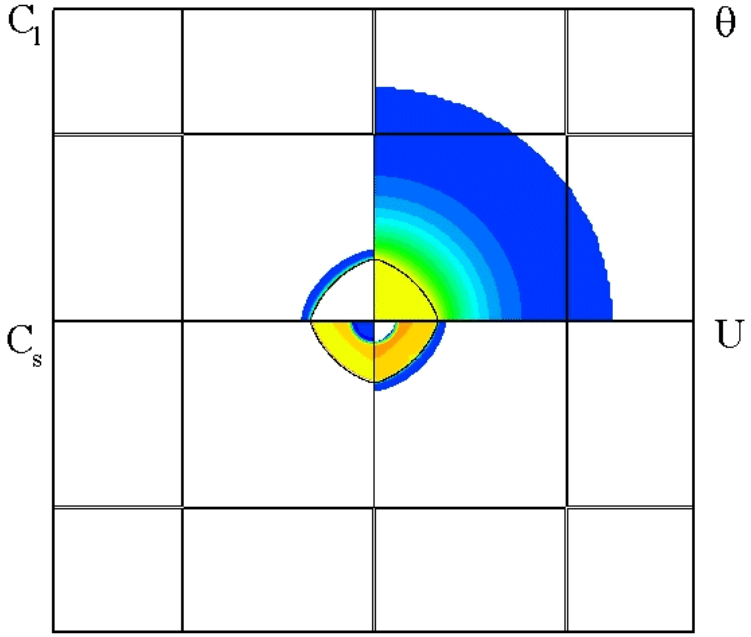
2-D Simulations of coupled columnar and equiaxed dendritic growth in directional solidification of a binary alloy



Badillo, A., and Beckermann, C., "Phase-Field Simulation of the Columnar-to-Equiaxed Transition in Alloy Solidification," Acta Materialia, Vol. 54, 2006, pp. 2015-2026.

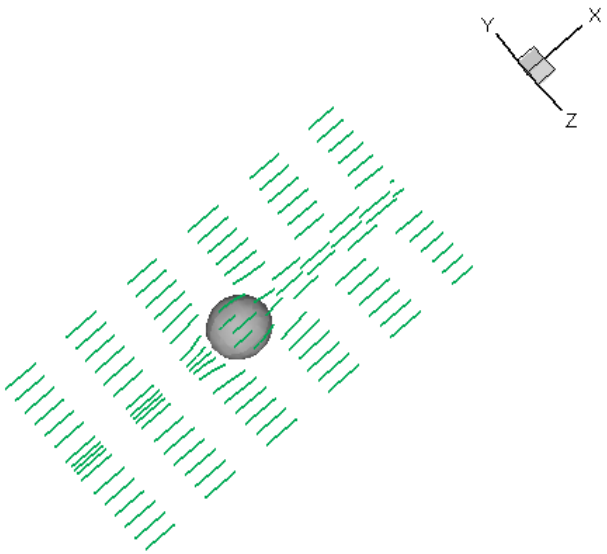
2-D Simulations of free dendritic growth of a dilute binary alloy with coupled heat and solute diffusion





Ramirez, J.C., Beckermann, C., Karma, A., and Diepers, H.-J., "Phase-Field Modeling of Binary Alloy Solidification with Coupled Heat and Solute Diffusion," Physical Review E, Vol. 69, 051607, 2004.

3-D Simulations of dendritic growth of a pure substance with flow in the melt



Lu, Y., Beckermann, C., and Ramirez, J.C., "Three-Dimensional Phase-Field Simulations of the Effect of Convection on Free Dendritic Growth," J. Crystal Growth, Vol. 280, pp. 320-334, 2005.

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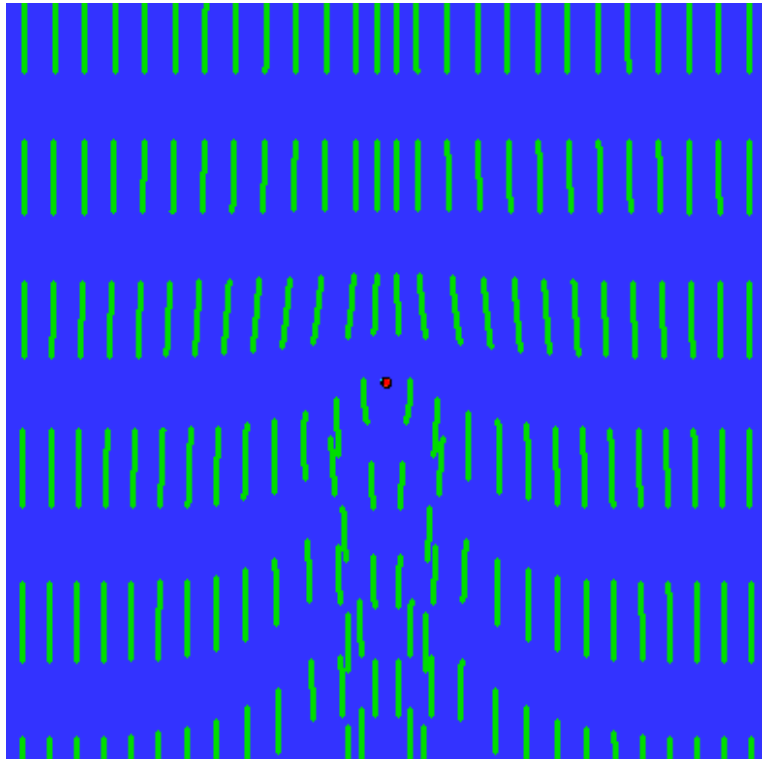

Tong, X., Beckermann, C., Karma, A. and Li, Q., "Phase-Field Simulations of Dendritic Crystal Growth in a Forced Flow," Physical Review E, Vol. 63, 061601, 2001.